ModelIt!

Student Activity Pack

Why Are Species Going Extinct So Fast?

Modeling Biodiversity Loss, Habitat Fragmentation, and Extinction Cascades

Grade 12L2 | NGSS HS-LS2-7 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 How many species do you think go extinct every day? What do you think is the main cause?

2 What is the difference between a species losing half of its habitat in one large area versus losing that same amount but split into many small, isolated fragments? Which is worse for the species?

3 Can you think of examples where the loss of one species could cause other species to decline? How might species be connected?

4 Why should humans care about biodiversity loss? What do we depend on from other species?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Land Use Change Rate | Climate Velocity | Habitat Connectivity | Population Viability | Species Interaction Network | Ecosystem Function Index | Extinction Rate

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Continued Habitat Loss

Land Use Change: Current Rate | Climate Velocity: Moderate | Habitat Connectivity: Declining

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Climate Without Corridors

Land Use Change: Current | Climate Velocity: High | Habitat Connectivity: Low (Fragmented)

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Connected Landscape

Land Use Change: Current | Climate Velocity: Moderate | Habitat Connectivity: High (Corridors)

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Invasive Species Pressure — The impact of non-native species introduced by human activity — invasive species compete with, prey on, or bring diseases to native species, and their impact is amplified in fragmented habitats where native populations are already stressed and cannot escape

Genetic Diversity Index — The range of genetic variation within a species population — small, isolated populations lose genetic diversity through drift and inbreeding, reducing their ability to adapt to changing conditions and making them vulnerable to diseases that can sweep through genetically uniform populations

Protected Area Effectiveness — The degree to which designated conservation areas actually protect biodiversity — many protected areas are too small, poorly managed, or isolated to maintain viable populations, and some exist only on paper with no real enforcement of protection

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Species are disappearing at 1,000 times the natural background rate — a pace not seen since the asteroid that killed the dinosaurs 66 million years ago — so what is driving this sixth mass extinction, and what happens to ecosystems and human civilization when the web of life unravels? Use evidence from your simulation to support your answers.

1. **Why does your model show that extinction rates accelerate non-linearly as habitat loss increases? What happens at the threshold where populations become too small to be viable?**

2. **How does habitat fragmentation multiply the damage of habitat loss? Use specific examples from your model to explain why connectivity matters more than total area.**

3. **What are extinction cascades and why do they mean the true impact of losing one species is always greater than it appears? Trace a cascade through your model\'s species interaction network.**

4. **Your model showed that wildlife corridors significantly reduce extinction rates even without stopping habitat loss. What is it about connectivity that makes this possible?**

5. **What are the ethical dimensions of biodiversity conservation — do species have a right to exist independent of their utility to humans, or is the ecosystem services argument sufficient justification for conservation?**

8 STEM Challenge: Design a Biodiversity Conservation Network

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YOUR ENGINEERING MISSION

Design a connected conservation network for a region experiencing rapid development that maximizes species survival by linking existing protected areas through strategic wildlife corridors.

THE SCENARIO

A region with exceptional biodiversity has 15 existing protected areas covering 12% of the landscape, but they are isolated from each other by agriculture, roads, and urban development. Climate models project that many species will need to shift their ranges 50-100 km northward over the next 50 years. Currently, the fragmented landscape makes this migration impossible for most species. A conservation coalition has hired your team to design a corridor network that connects the 15 protected areas, enables climate migration, and prioritizes the species and habitats most at risk. Your budget is limited, so you must use your model data to identify the most strategic connections.

Engineering Design Thinking

1. WHICH CORRIDOR CONNECTIONS DOES YOUR NETWORK PRIORITIZE AND WHY — WHAT MODEL EVIDENCE SUPPORTS THESE CHOICES?

2. HOW DOES YOUR NETWORK DESIGN ACCOUNT FOR BOTH CURRENT BIODIVERSITY HOTSPOTS AND PROJECTED CLIMATE MIGRATION NEEDS?

3. WHAT KEYSTONE SPECIES DOES YOUR PLAN SPECIFICALLY TARGET FOR PROTECTION, AND HOW DOES PREVENTING THEIR LOSS REDUCE CASCADE RISK?

4. HOW DOES YOUR PLAN BALANCE CONSERVATION NEEDS WITH THE ECONOMIC INTERESTS OF LANDOWNERS AND DEVELOPERS IN CORRIDOR ZONES?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

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Explain your design using scientific evidence from the simulation:

Real Career Connection

Conservation biologists who study biodiversity loss and design protected area networks work for national parks, wildlife agencies, and organizations like the Nature Conservancy and World Wildlife Fund, earning $50,000-$110,000/year. Wildlife corridor planners who design landscape-level connectivity for species movement earn $55,000-$105,000/year. Ecological modelers who build computational simulations of species interactions and extinction dynamics earn $65,000-$130,000/year at universities and research institutions.